

MTH209: Worksheet 13

Penalized Regression

Ridge regression

The ridge regression estimator is defined, for a user-chosen $\lambda \geq 0$ as,

$$\hat{\beta}_{\text{OLS}} = \arg \min_{\beta \in \mathbb{R}^p} \{(y - X\beta)^\top (y - X\beta) + \lambda \beta^\top \beta\}.$$

It is an homework to show that in this case, there is a closed-form solution:

$$\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I_p)^{-1} X^\top y,$$

where $\lambda \geq 0$ is a regularization parameter that controls the amount of shrinkage/bias. Equivalently, it can be viewed as the solution to the constrained problem

$$\min_{\beta} \|y - X\beta\|^2 \quad \text{subject to } \|\beta\|^2 \leq t,$$

for some $t > 0$. Large λ corresponds to a tighter constraint (smaller t). Further,

- As $\lambda \rightarrow 0$, $\hat{\beta}_{\text{ridge}} \rightarrow \hat{\beta}_{\text{OLS}}$.
- As $\lambda \rightarrow \infty$, $\hat{\beta}_{\text{ridge}} \rightarrow 0$.

Ridge regression thus introduces bias but is useful because it can substantially reduces variance, particularly in directions where X is ill-conditioned. The overall mean squared error (MSE) can therefore be smaller than that of the unbiased OLS estimator. We will see this in a simulation.

The Lasso and Sparsity in Linear Models

The Lasso (Least Absolute Shrinkage and Selection Operator) is a regularization method for linear regression that performs both coefficient shrinkage and variable selection.

Consider the linear model

$$y = X\beta + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2 I_n),$$

where X is an $n \times p$ design matrix and β is a vector of coefficients. The Lasso estimator is defined as

$$\hat{\beta}_{\text{lasso}} = \arg \min_{\beta} \left(\|y - X\beta\|^2 + \lambda \|\beta\|_1 \right),$$

where $\|\beta\|_1 = \sum_{j=1}^p |\beta_j|$ and $\lambda \geq 0$ is a tuning parameter controlling sparsity.

Equivalently, it can be written as the constrained problem

$$\min_{\beta} \|y - X\beta\|^2 \quad \text{subject to } \|\beta\|_1 \leq \sqrt{t},$$

for some $t > 0$ corresponding to λ . The Lasso differs from ridge regression in the geometry of its constraint set. The ℓ_1 -ball $\{\beta : \|\beta\|_1 \leq t\}$ has corners aligned with the coordinate axes, while the ridge (ℓ_2) constraint $\{\beta : \|\beta\|_2 \leq t\}$ is smooth and circular. Contours of the least-squares objective are ellipsoids centered at the ordinary least squares (OLS) solution.

The Lasso solution occurs where these ellipsoids first touch the ℓ_1 -ball, which often happens at a corner, **forcing some coefficients to be exactly zero**.

Unlike ridge regression, the Lasso has no closed-form solution. But geometrically, the Lasso introduces bias even for large signal coefficients, since the ℓ_1 penalty shrinks all coefficients by a constant amount rather than proportionally. However, by eliminating irrelevant predictors, the Lasso often achieves a lower prediction variance than OLS or ridge regression in high-dimensional settings.

- The ℓ_1 penalty yields sparse models by pushing some coefficients exactly to zero.
- The resulting estimator trades additional bias for potentially large variance reduction when p is large or predictors are correlated.
- When predictors are highly correlated, Lasso tends to select one variable from each correlated group.
- Ridge regression, by contrast, never sets coefficients exactly to zero but distributes shrinkage smoothly across correlated directions.

Questions (only on Ridge)

1. Let us first simulate fake data according to the linear regression model $y = X\beta + \epsilon$. This can be done by fixing β and generating a random X and then drawing the response y from a normal.

```
set.seed(1)
n <- 50
p <- 5
beta.star <- c(-.5, .3, 1, 5, -2)

# Making design matrix
X <- matrix(rnorm(n*p), nrow = n, ncol = p)

# Generating response
y <- X %*% beta.star + rnorm(n, mean = 0, sd = 1)
```

- a. Calculate the OLS estimator of β .
 - b. For $\lambda = 1$ calculate the Ridge estimator of β .
 - c. For $\lambda = 10$ calculate the Ridge estimator of β .
 - d. What do you observe in b and c?
2. We discussed that ridge regression estimator is biased. Let us “verify” this. First, let us simulate a 1-dimensional problem:

```
n <- 50
p <- 1
beta.star <- c(1) # btrue beta is 1

# Making design matrix
X <- matrix(rnorm(n*p), nrow = n, ncol = p)

# Generating response
y <- X %*% beta.star + rnorm(n, mean = 0, sd = 1)
```

- a. Set up a simulation study to verify that OLS is unbiased.
 - b. For $\lambda = 10$ set up a simulation study to verify that ridge regression is unbiased.
3. Repeat the exercise in 2. to conclude that ridge regression estimator of β has smaller variance.
4. Finally, set up a simulation study to calculate the mean squared error of OLS and Ridge regression estimators for various values of λ and draw a plot of the MSE of Ridge versus λ , along with the OLS MSE as a horizontal line. What do you conclude?